Srikrishna Bangalore Raghu

Doctoral Student in Computer Science

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Summary

My research develops learning-based motion generation models deployed on real manipulator hardware. I am seeking a Summer 2027 robotics internship at the intersection of motion planning and robot learning.

Education

University of Colorado Boulder (Boulder, CO, USA) 2023–Present
Ph.D. in Computer Science (GPA 3.889/4) — Expected Aug 2028

PES University (Bengaluru, India) 2017–2021
B.Tech in Electrical and Electronics Engineering (CGPA 8.78/10)

Research Experience

PhD Student, HIRO Group, University of Colorado Boulder, Advisor: Alessandro Roncone Aug 2023–Present

- Developed a hesitancy-conditioned diffusion model (PyTorch) that generates real-world Franka Emika Panda trajectories at a commanded level of hesitancy.
- Built the collection and labeling pipeline for Dance2Hesitate, a 220-trajectory multimodal dataset of hesitant motion (synchronized 30 Hz robot-state streams from Franka kinesthetic teaching), released open source on Hugging Face. Link
- Trained Transformer and LSTM decoders that classify hesitancy intensity from motion (leave-one-dancer-out $\kappa = 0.76$ on unseen performers) and ran an in-person perception study ($N = 33$).
- Generated emotionally and functionally expressive manipulator trajectories using Laban Movement Analysis (LMA). Human trials supported 2 of 3 hypotheses. Link
- Designed and implemented BBoE, a bidirectional kinodynamic motion planner in Python, the only planner in our benchmark to reach $\geq 96\%$ success on all four systems (3D to 6D), with the lowest planning time on three.

Research Assistant, Artificial Intelligence and Robotics Lab, Indian Institute of Science Jul 2021–Jun 2022

- Implemented a heterogeneous multi-task multi-robot task allocation algorithm for TurtleBot3 systems, deployed in simulation and on real robots via ROS 2 with Gmapping SLAM.

Publications

- **S. Bangalore Raghu**, A. Soukhovei, D. Vankineni, A. Roncone. *What makes Robot Motion Hesitant? Producing and Perceiving Hesitancy in Robotic Manipulators*. **ACM/IEEE International Conference on Human-Robot Interaction (HRI)**, 2027. (*Submitted*).
- **S. Bangalore Raghu**, A. Soukhovei, D. Vankineni, A. Bacula, A. Roncone. *Dance2Hesitate: A Multi-Modal Dataset of Dancer-Taught Hesitancy for Understandable Robot Motion*. **Designing Transparent and Understandable Robots (D-TUR) Workshop at the International Conference on Human-Robot Interaction (HRI)**, 2026. Link
- **S. Bangalore Raghu**, A. Roncone. *BBoE: Leveraging a Bundle of Edges for Kinodynamic Bidirectional Motion Planning*. **Robotics and Automation Letters (RA-L)**, 2026. (*Accepted*).
- **S. Bangalore Raghu**, C. Lohrmann, A. Bakshi, J. Kim, J. C. Herrera, B. Hayes, A. Roncone. *Employing Laban Shape for Generating Emotionally and Functionally Expressive Trajectories in Robotic Manipulators*. **IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)**, 2025. Link
- S. B. Velhal, **S. Bangalore Raghu**, M. Bharatheesha, S. Sundaram. *A Dynamic Heterogeneous Team-based Non-iterative Approach for Online Pick-up and Just-In-Time Delivery Problems*. **arXiv**, 2024. (*Preprint*). arXiv

Awards & Teaching

Departmental Summer Research Fellowship, University of Colorado Boulder 2026
Published Work Award, Research Expo, University of Colorado Boulder 2026
Graduate Teaching Assistant (CSCI 1300, 2270, 3302; ROBO 5000), University of Colorado Boulder 2023–Present

Skills

Programming: Python, C++, C, MATLAB, Java, R, Julia **Hardware:** Franka Emika Panda, TurtleBot3
Robotics/ML: Motion Planning, Diffusion Models, Transformers, Deep Learning, Reinforcement Learning, HRI
Frameworks/Tools: PyTorch, ROS/ROS 2, Linux, MoveIt, libfranka, Gazebo, PyBullet, Webots, OpenCV, Docker, Git, AWS